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| <p><b>5</b></p> <p><b>Statistical hypothesis testing</b></p> | <p>5.1</p> | <p><b>Understand and apply the language of statistical hypothesis testing, developed through a binomial model: null hypothesis, alternative hypothesis, significance level, test statistic, 1-tail test, 2-tail test, critical value, critical region, acceptance region, <math>p</math>-value;</b></p> <p>extend to correlation coefficients as measures of how close data points lie to a straight line.</p> <p>and</p> <p>be able to interpret a given correlation coefficient using a given <math>p</math>-value or critical value (calculation of correlation coefficients is excluded).</p> | <p><b>An informal appreciation that the expected value of a binomial distribution is given by <math>np</math> may be required for a 2-tail test.</b></p> <p>Students should know that the product moment correlation coefficient <math>r</math> satisfies <math> r  \leq 1</math> and that a value of <math>r = \pm 1</math> means the data points all lie on a straight line.</p> <p>Students will be expected to calculate a value of <math>r</math> using their calculator but use of the formula is not required.</p> <p>Hypotheses should be stated in terms of <math>\rho</math> with a null hypothesis of <math>\rho = 0</math> where <math>\rho</math> represents the population correlation coefficient.</p> <p>Tables of critical values or a <math>p</math>-value will be given.</p> |
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| Topics                                                                | What students need to learn: |                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|                                                                       | Content                      |                                                                                                                                                                                                                                                                                                                                                                    | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>5</b><br><b>Statistical hypothesis testing</b><br><i>continued</i> | 5.2                          | <p><b>Conduct a statistical hypothesis test for the proportion in the binomial distribution and interpret the results in context.</b></p> <p><b>Understand that a sample is being used to make an inference about the population and</b></p> <p><b>appreciate that the significance level is the probability of incorrectly rejecting the null hypothesis.</b></p> | <p><b>Hypotheses should be expressed in terms of the population parameter <math>p</math></b></p> <p><b>A formal understanding of Type I errors is not expected.</b></p>                                                                                                                                                                                                                                                                                                                            |
|                                                                       | 5.3                          | <p>Conduct a statistical hypothesis test for the mean of a Normal distribution with known, given or assumed variance and interpret the results in context.</p>                                                                                                                                                                                                     | <p>Students should know that:</p> <p>If <math>X \sim N(\mu, \sigma^2)</math> then <math>\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)</math> and that a test for <math>\mu</math> can be carried out using:</p> $\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1^2).$ <p>No proofs required.</p> <p>Hypotheses should be stated in terms of the population mean <math>\mu</math>.</p> <p>Knowledge of the Central Limit Theorem or other large sample approximations is not required.</p> |